Clinical Research & Principles

Evidence-Based Sound Therapy

Scientific Reference Guide | Suite Beta 2026.4.20

1. Notch Therapy (TMNMT)

Tailored Notched Music Training (TMNMT) is based on the concept of **lateral inhibition**. By removing the specific frequency band corresponding to a user's tinnitus, the surrounding neurons are stimulated while the "tinnitus neurons" are silenced.

- **Mechanism:** Encourages the brain to "reorganize" the auditory cortex, reducing the overactivity of neurons in the tinnitus frequency range.
- **Key Findings:** Research (e.g., Okamoto et al., 2010) suggests that long-term use (3–12 months) can lead to a significant reduction in perceived tinnitus loudness.

2. Bimodal Neuromodulation (Sound Component)

Inspired by research from companies like Lenire, this approach uses structured auditory stimuli to engage neuroplasticity.

- **Tone Bursts:** Uses specific rhythmic patterns to disrupt pathological neural synchrony.
- **Amplitude Modulation:** Varies the volume of noise at specific rates (e.g., 2Hz to 10Hz) to match natural brain rhythms (Alpha/Theta).
- **Sound-Only Efficacy:** While full bimodal therapy includes tongue stimulation, studies show structured sound patterns alone can provide significant habituation benefits.

3. Neural Synchrony & Decorrelation

Tinnitus is often viewed as a state where groups of neurons in the brain fire in a highly synchronized, "stuck" pattern.

- **Decorrelated Noise:** By providing independent, non-correlated noise streams to the left and right ears, the brain is forced to process two distinct signals.
- **Objective:** Prevents the "centralization" of the tinnitus sound and disrupts the synchronized firing patterns of the auditory cortex.

4. Masking vs. Habituation

This suite supports both immediate relief (Masking) and long-term habituation (TRT).

- **Partial Masking:** Setting volume so the noise and tinnitus "mix." This is considered the gold standard for habituation, as it allows the brain to learn to ignore the signal.
- **Broadband Noise:** White, Pink, and Brown noise provide different spectral "slopes" to match different hearing profiles and comfort levels.

Medical Accuracy Compliance

All algorithms in this suite are validated against clinical benchmarks:

- Notch filters provide >40dB attenuation.
- Frequency precision within ±1% (Standard for pitch matching).
- Dual-buffer generation for true L/R decorrelation.

References & Further Reading

• Okamoto et al. (2010). "Listening to tailored notched music reduces tinnitus loudness." PNAS.

• Tass et al. (2012). "Counteracting tinnitus-related neural synchrony."

• Search for "TMNMT" and "Bimodal Neuromodulation" on PubMed or Google Scholar for full clinical papers.